

YOLOv8 Real-Time Object Detection – Architecture & Flowchart

System Architecture

Webcam → OpenCV Frame Capture → YOLOv8 Model → Object Detection → Bounding Box & Label Rendering (CVZone) → FPS Calculation → Display Output

Architecture Explanation

- Webcam captures live video frames using OpenCV.
- Each frame is passed to YOLOv8 for object detection.
- YOLOv8 returns bounding boxes, confidence scores, and class IDs.
- CVZone draws stylish bounding boxes and labels.
- FPS is calculated using frame timestamps.
- Processed frames are displayed in real time.

Process Flowchart

Start → Initialize Webcam & Model → Capture Frame → Run YOLOv8 Detection → Extract Boxes → Draw Bounding Boxes & Labels → Calculate FPS → Display Frame → Repeat → End

Conclusion

This document explains the complete workflow and architecture of a YOLOv8-based real-time object detection system using Python, OpenCV, and CVZone.